

Rising Waters: A Machine Learning Approach to Flood Prediction

Performance Testing

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Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Flask Testing Environment
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Flood Prediction Module
Test Environment	Local System (Flask Application)
Test Date	30 june 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single Flood Prediction Request	1	30	Prediction generated successfully
2	Multiple Consecutive Predictions	20	60	Stable performance without errors
3	High Rainfall Data Processing	50	120	Accurate flood classification
4	Peak Load Prediction Requests	100	180	System remains responsive

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.85 sec	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	2.10 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	-	145 Req/sec	Pass	Stable request handling
4	Error Rate	< 1%	0.2%	Pass	Minimal errors observed
5	CPU Utilization	< 80%	52%	Pass	Efficient resource usage
6	Memory Utilization	< 80%	48%	Pass	Stable memory consumption

Step 4: Observations & Analysis

The Flood Prediction System demonstrated stable performance under both normal and peak load conditions. Prediction requests were processed efficiently with low response times and minimal error rates. Resource utilization remained within acceptable limits, confirming the suitability of the application for real-time flood risk assessment.